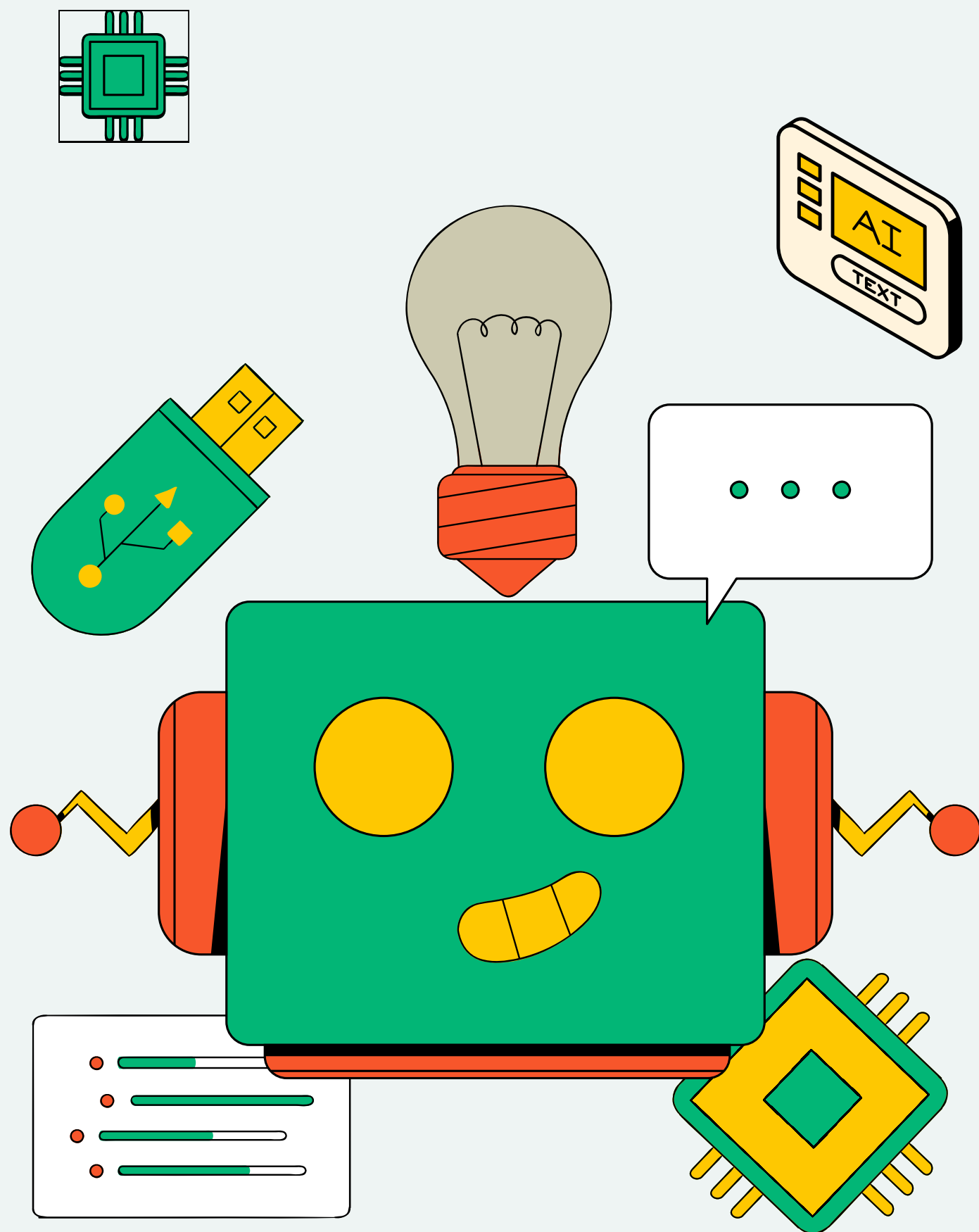


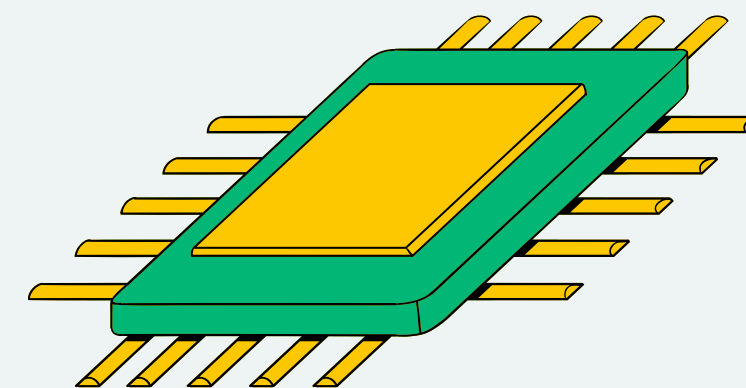


MARKET BASKET ANALYSIS: AN MLOPS APPROACH PRESENTATION

PRESENTED BY:

ZERGUERRAS KHAYRA SARRA

FEZAZI AMINA KHADIDJA



PROJECT OBJECTIVE



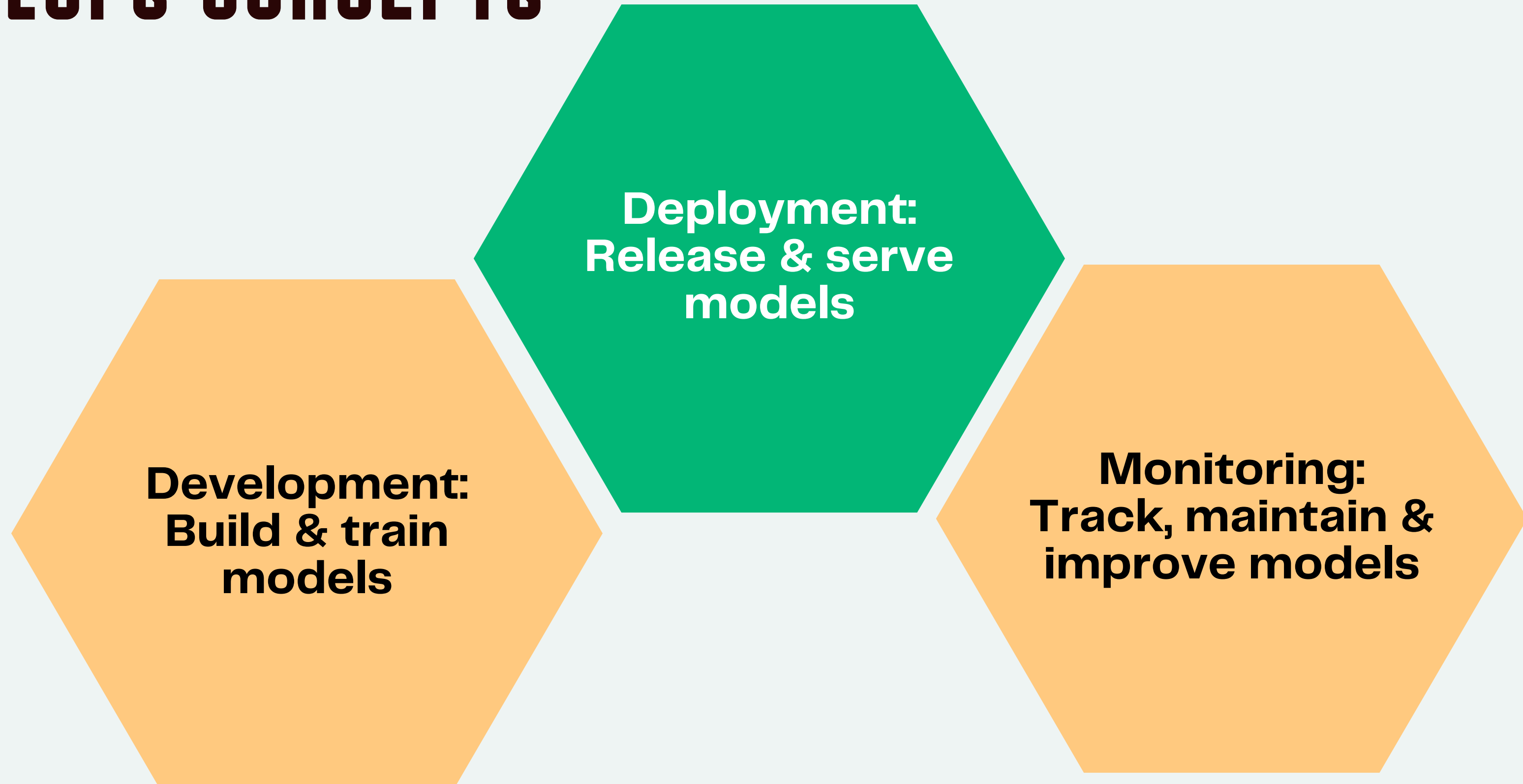
- Gain hands-on experience with Weights & Biases (W&B) as an MLOps platform
- Build an end-to-end Market Basket Analysis pipeline from raw data to recommendations
- Compare Apriori and FP-Growth algorithms with systematic experiment tracking
- Apply MLOps practices (artifact versioning, experiment comparison) for reproducibility
- Create a rule-based recommendation system demonstrating practical business value

WHAT IS MLOPS?

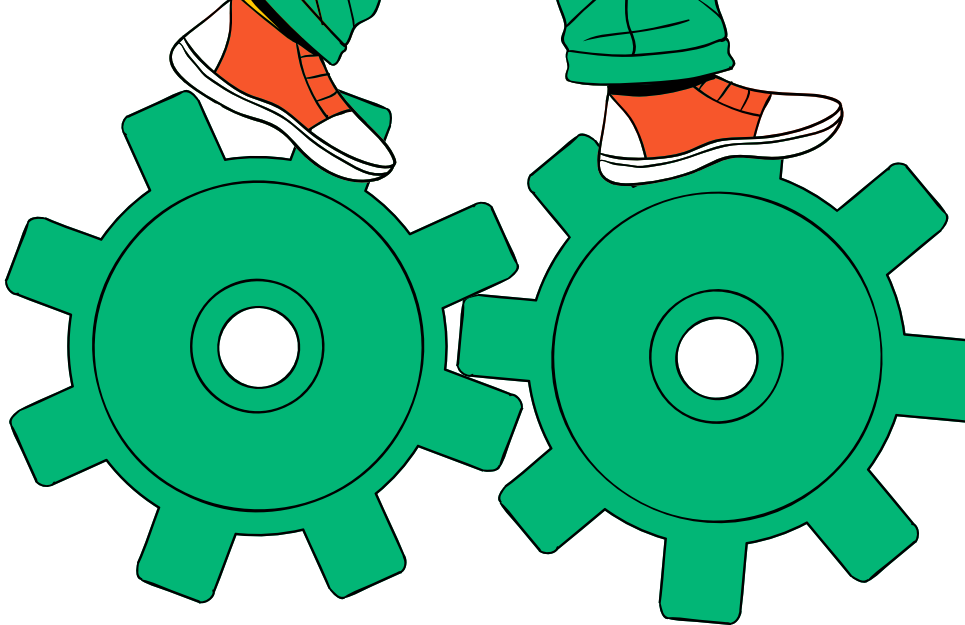
MLOps (Machine Learning Operations) is a set of practices that combines machine learning and DevOps principles to manage the entire ML lifecycle. It ensures that ML models are reproducible, scalable, and deployable, enabling efficient collaboration and reliable performance from experimentation to production.



MLOPS CONCEPTS



WHY DID WE CHOOSE WEIGHTS & BIASES ?



Experiment Tracking: Track experiments, including parameters, metrics, and artifacts

ak-fezazi-university > Projects > instacart-market-basket-... > Table

Ak-fezazi's workspace

Personal workspace

Saved just now

Runs 425

Search runs

Filter Group Sort New sweep

NAME	425 visualized	STATE	NOTES	USER	TAGS	CREATED	RUNTIME	SWEEP	ALGORITHM	MIN_CONFIDENCE	MIN_LIFT	MIN_SUPPORT
Apriori_..._run16		Finished	Add notes...	ak-fezazi		4w ago	2s	-	Apriori	0.3	2.41246	0.01
Apriori_..._run15		Finished	Add notes...	ak-fezazi		4w ago	1s					
Apriori_..._run14		Finished	Add notes...	ak-fezazi		4w ago	1s					
Apriori_..._run13		Finished	Add notes...	ak-fezazi		4w ago	1s					
Apriori_..._run12		Finished	Add notes...	ak-fezazi		4w ago	1s					
Apriori_..._run11		Finished	Add notes...	ak-fezazi		4w ago	2s					
Apriori_..._run10		Finished	Add notes...	ak-fezazi		4w ago	1s					

ak-fezazi-university > Projects > instacart-market-basket-... > Artifacts

FEZAZI AMINA KHADIDJA

ak-fezazi-university-org

Artifacts

Find matching artifacts

instacart-transaction-matrix-updated

Artifact overview

Type dataset

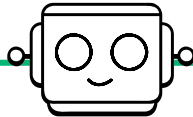
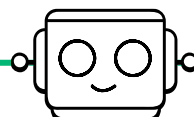
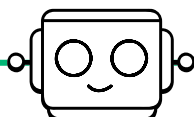
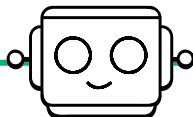
Created At December 30th, 2025

Description

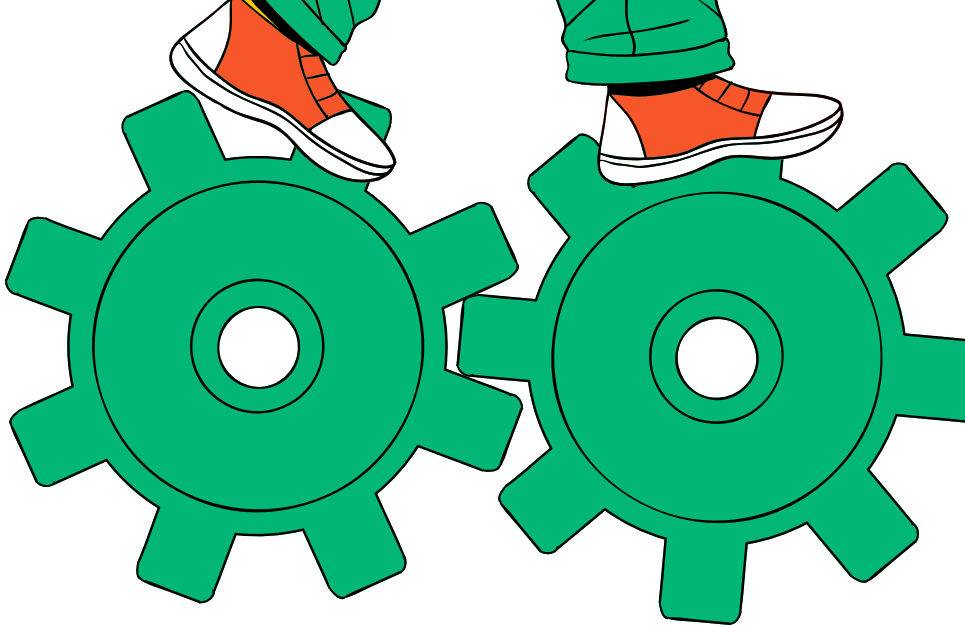
Versions

Show run metrics

Version	Aliases	Logged By	Tags	Created	TTL Remain...	# of Consu...	Size	m.new_sam...	m.update_t...	m.total_sa...	m.simulate...
0	@ latest	DATA_UP...		Tue Dec 30 2025	Inactive	0	309.9kB	27361	incr...	77361	True



WHY DID WE CHOOSE WEIGHTS & BIASES ?



Model Versioning: Manage and version models in a centralized, organized repository

Artifacts

Find matching artifacts

dataset

instacart-transaction-m...

v0 latest

best_association_model

instacart-transaction-m...

instacart-transactions-p...

model

retrained-association-m...

v0 latest retrained_v2

best-association-rules-...

v0 best latest production

documentation

project-documentation

rules

apriori-rules-ddd0a667

best-association-rules-model

Versions

Artifact overview

Type

model

Created At

December 30th, 2025

Description

Versions

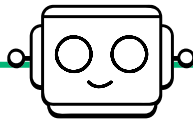
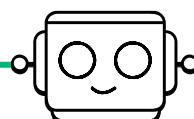
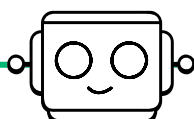
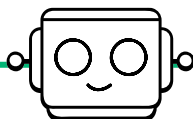
Show run metrics

1-1 of 1

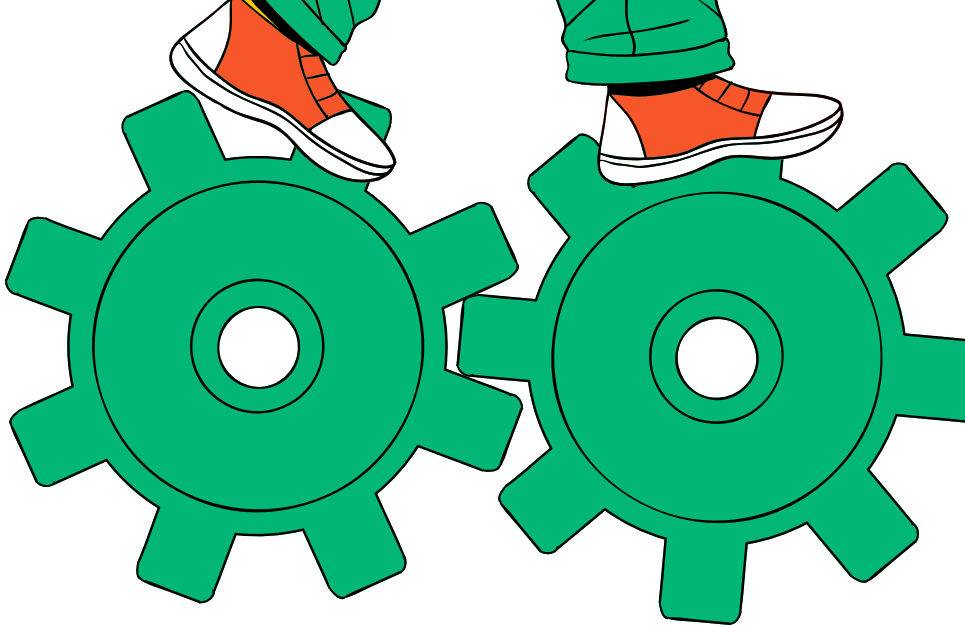
<

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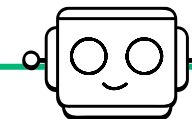
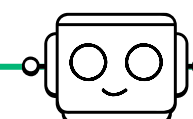
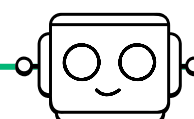
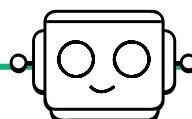
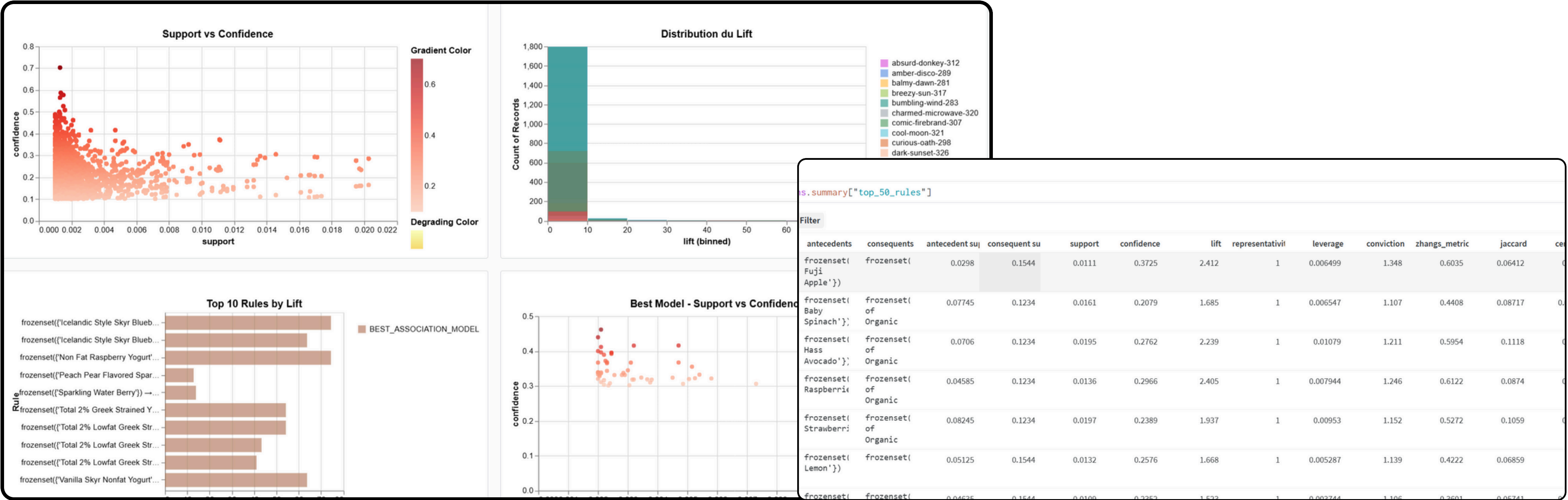
m.n_rules	m.avg_lift	m.max_lift	m.min_lift	m.algorithm	m.model_ty...	m.rules_path	m.min_sup...	m.runtime_...	m.total_rules	m.norm_n_...	m.norm_ru...	m.norm_av..
0	11.486	74.442	1	apriori	associa	apriori_	0.002	4.656	54	0	0.2646	0.8631



WHY DID WE CHOOSE WEIGHTS & BIASES?



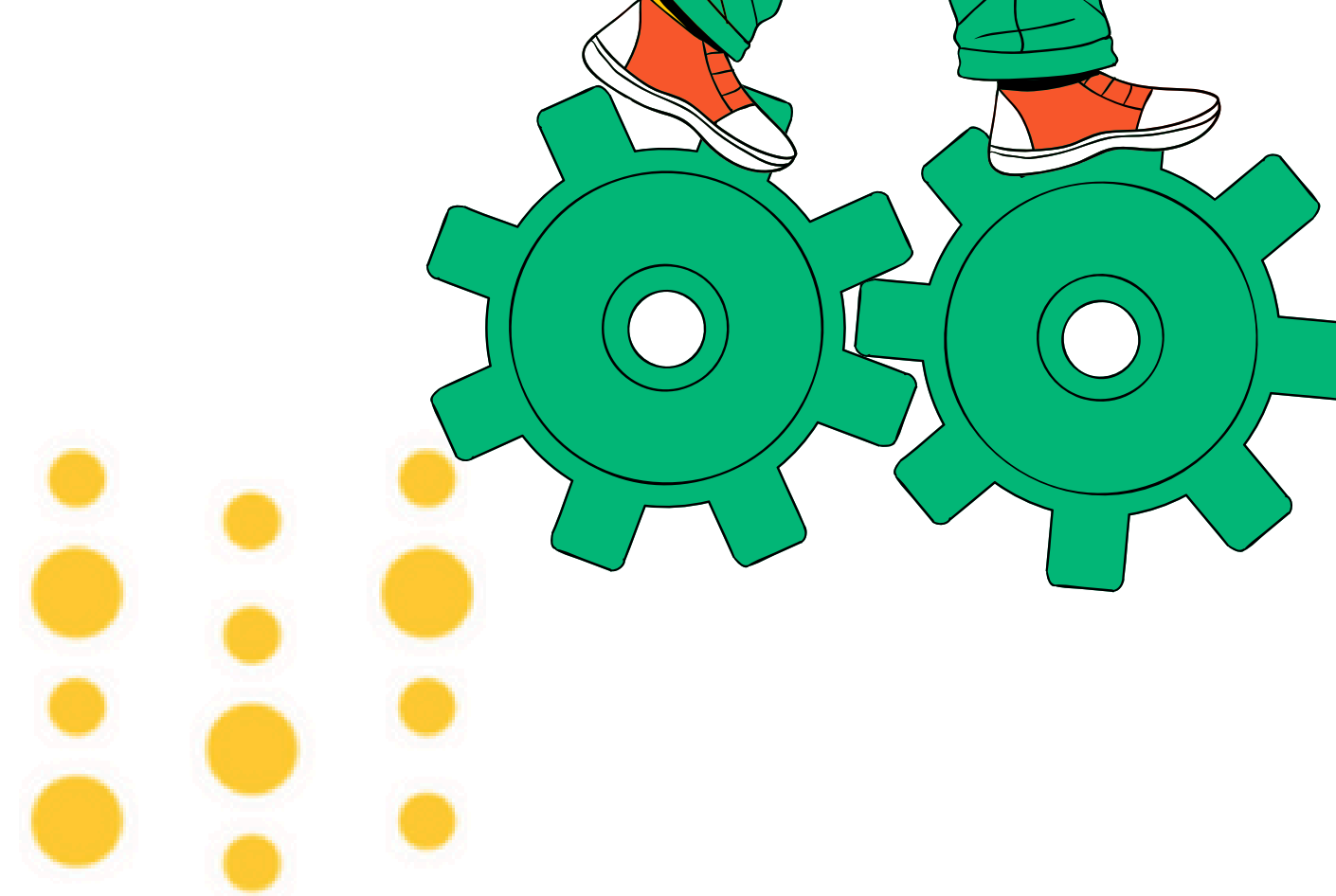
Collaboration & Visualization: Share insights, visualize results, and foster teamwork



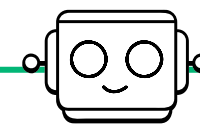
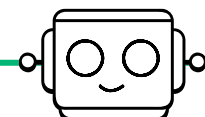
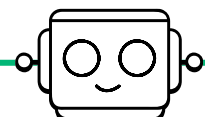
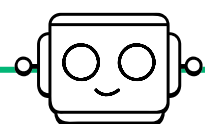
WHY DID WE CHOOSE WEIGHTS & BIASES ?

Library Agnostic: Compatible with any ML framework (TensorFlow, PyTorch, etc.)

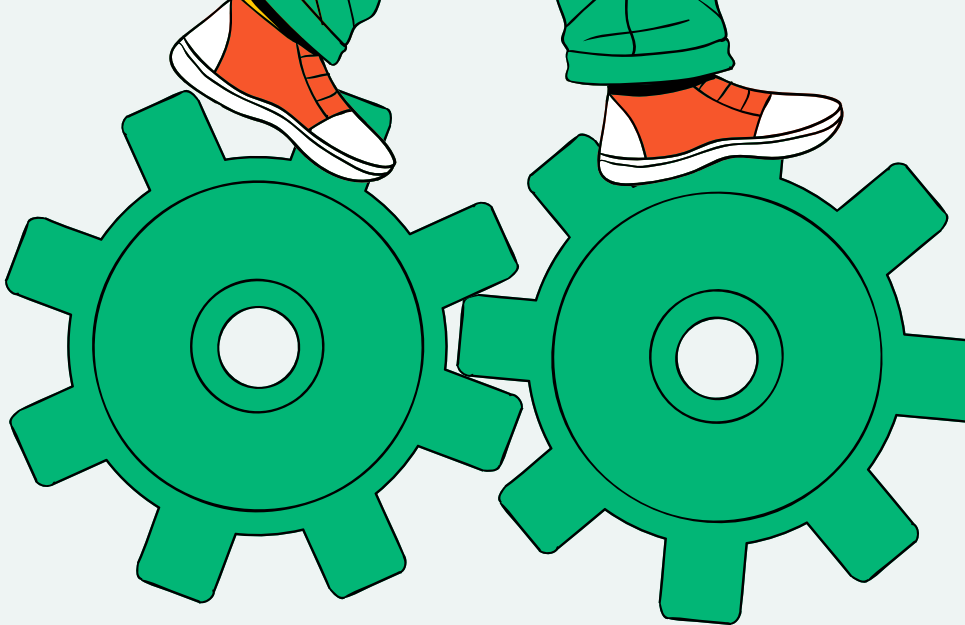
Scalable & Open Source: Extensible, community-driven, and freely available



Weights & Biases



DTASET OVERVIEW

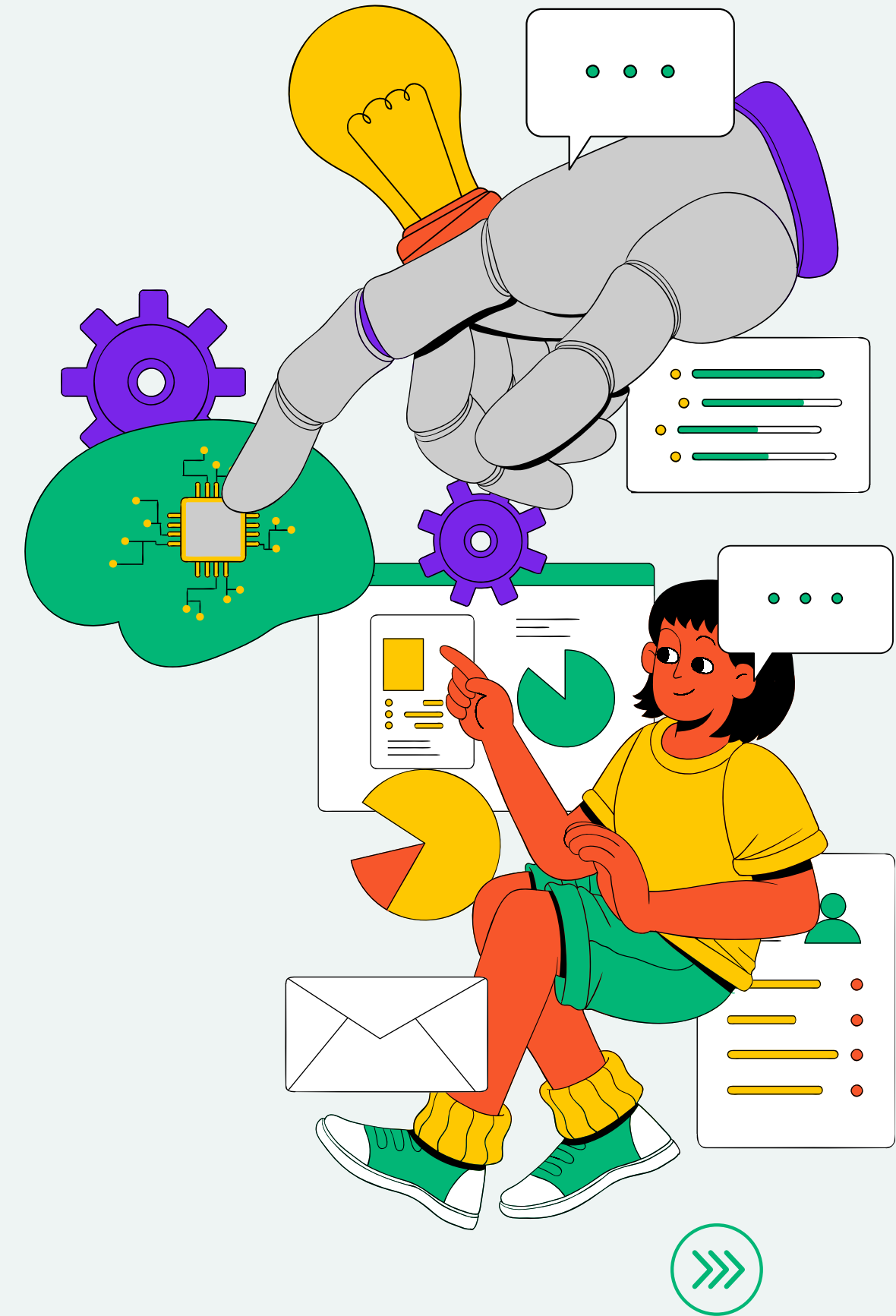


The project uses the Instacart Market Basket Analysis dataset, which consists of multiple relational files describing customer orders and purchased products. Order-level information is provided in orders.csv, while order_products__prior.csv contains the products included in each order and serves as the core table for building transactional baskets. Product metadata is enriched using products.csv, aisles.csv, and departments.csv, enabling the association of products with meaningful categories and supporting effective market basket analysis.

	order_id	product_id	add_to_cart_order	reordered	product_name	aisle_id	department_id	aisle	department	user_id	order_dow	order_hour_of_day	days_since_prior_order
0	2	33120	1	1	Organic Egg Whites	86	16	eggs	dairy eggs	202279	5	9	8.0
1	2	28985	2	1	Michigan Organic Kale	83	4	fresh vegetables	produce	202279	5	9	8.0
2	2	9327	3	0	Garlic Powder	104	13	spices seasonings	pantry	202279	5	9	8.0
3	2	45918	4	1	Coconut Butter	19	13	oils vinegars	pantry	202279	5	9	8.0
4	2	30035	5	0	Natural Sweetener	17	13	baking ingredients	pantry	202279	5	9	8.0

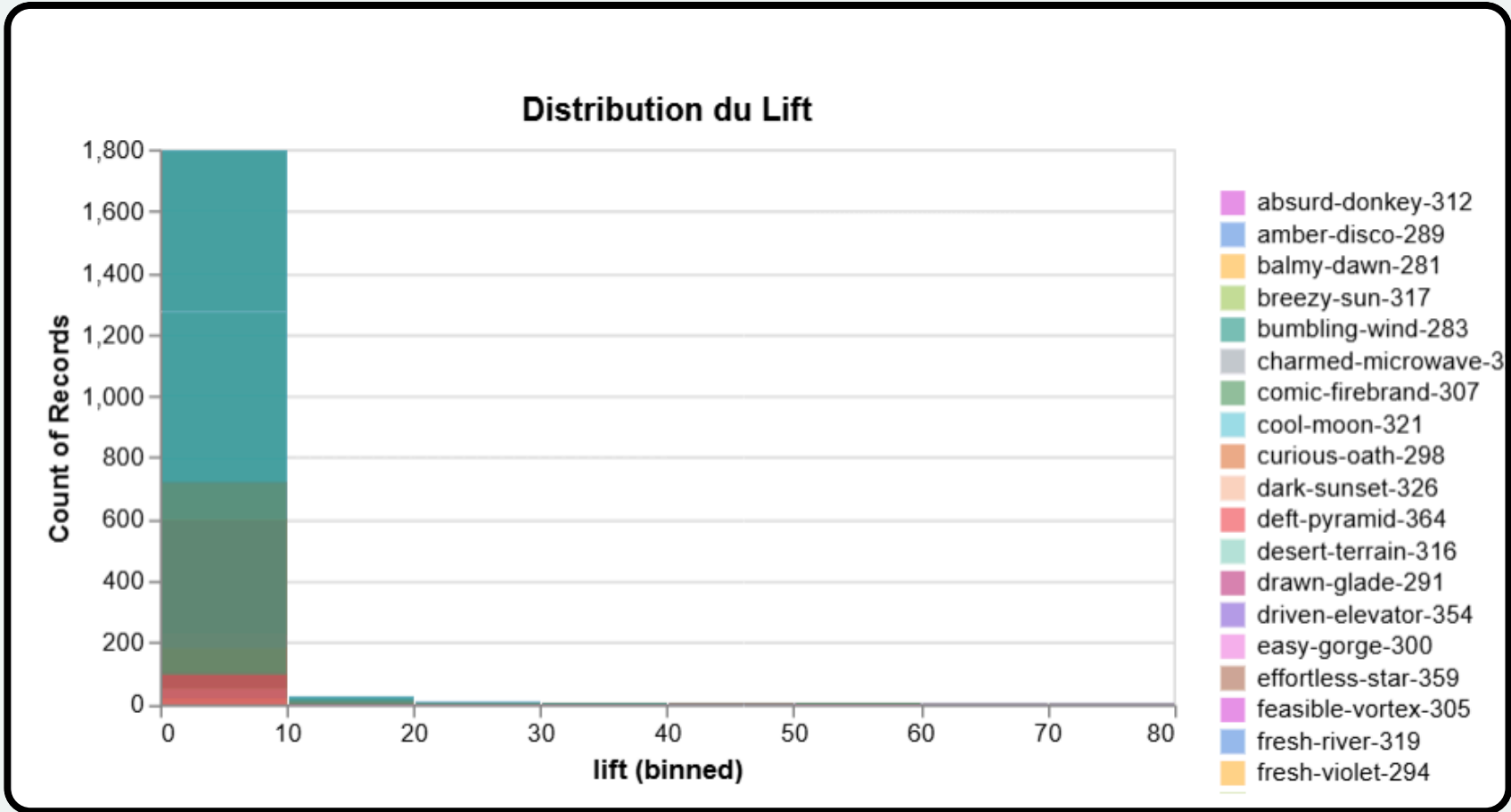
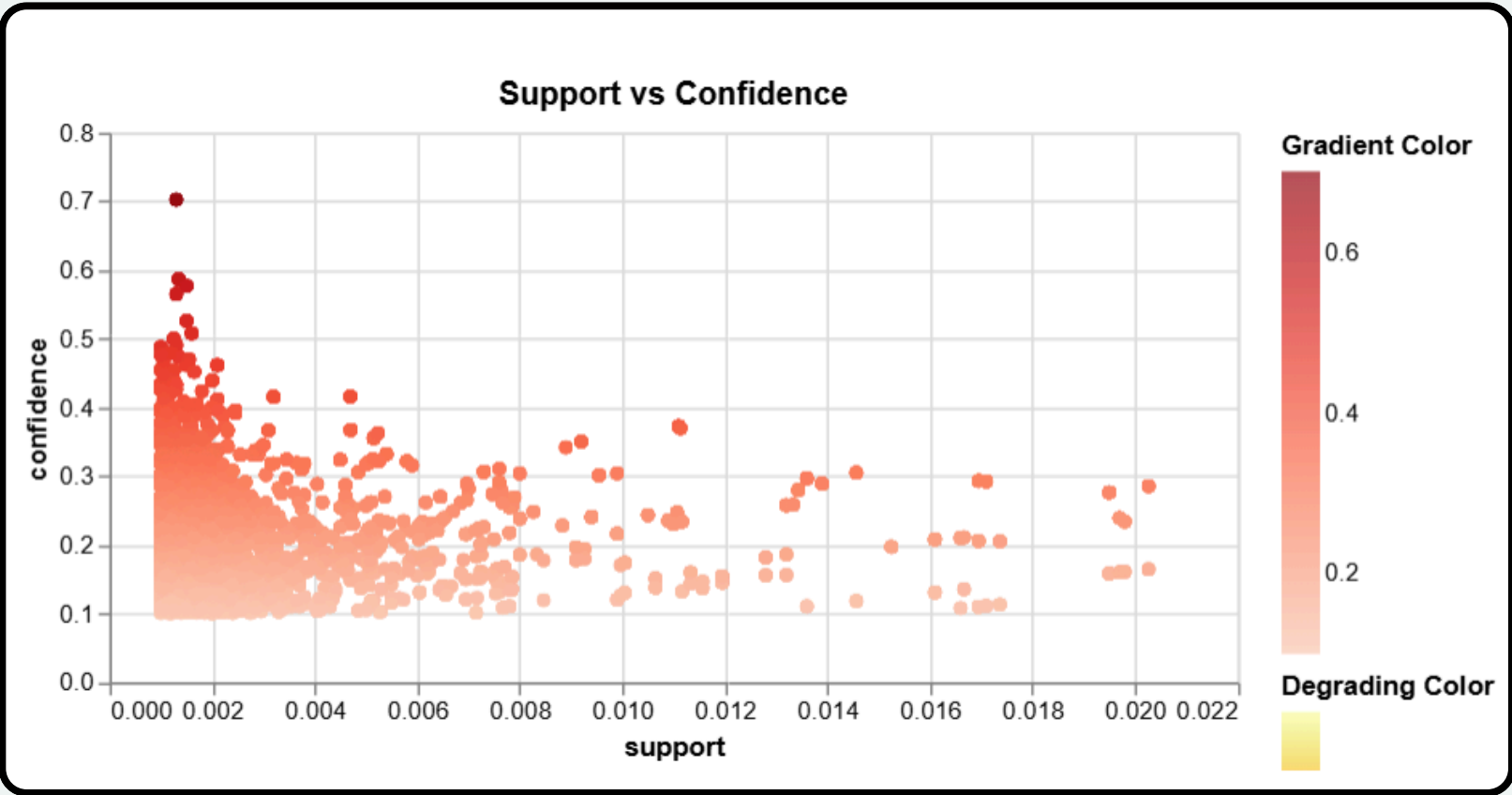
APPROACH

This project uses Market Basket Analysis (MBA) to uncover associations between products in transactional data. We evaluated two classical algorithms: **Apriori**, which systematically explores itemsets and is highly interpretable but can be computationally expensive, and **FP-Growth**, which compresses data into a compact tree structure, offering better scalability on large datasets. These algorithms are compared using standard metrics — support, confidence, and lift — to identify meaningful and actionable association rules.



COMPARAISON BETWEEN THE ALGORITHMES

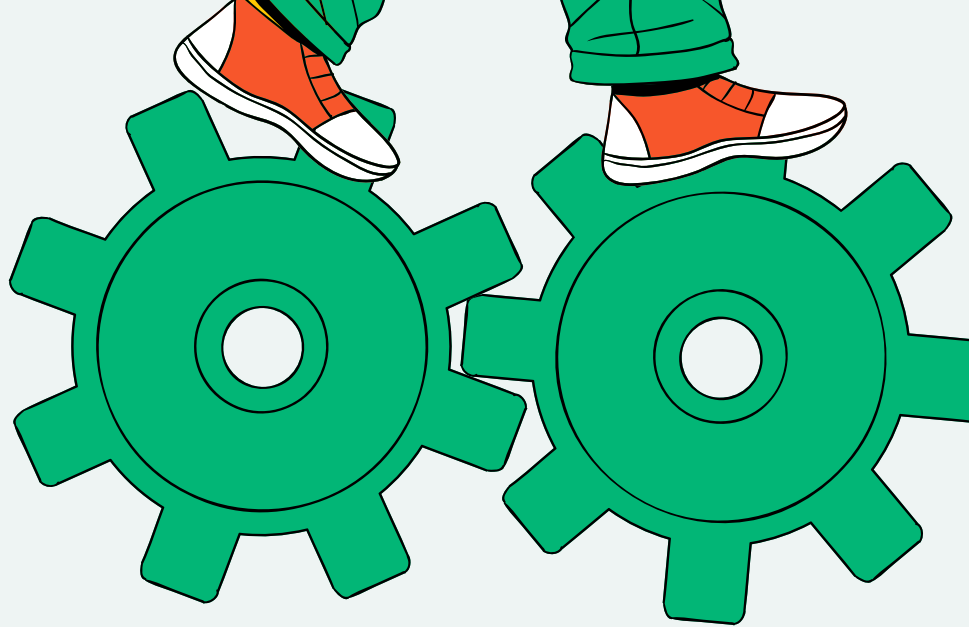
The support and confidence comparaison in the Apriori and Fp_Growth



The distrubution of the lift between the algorithmes



ALGORITHM PERFORMANCE ANALYSIS

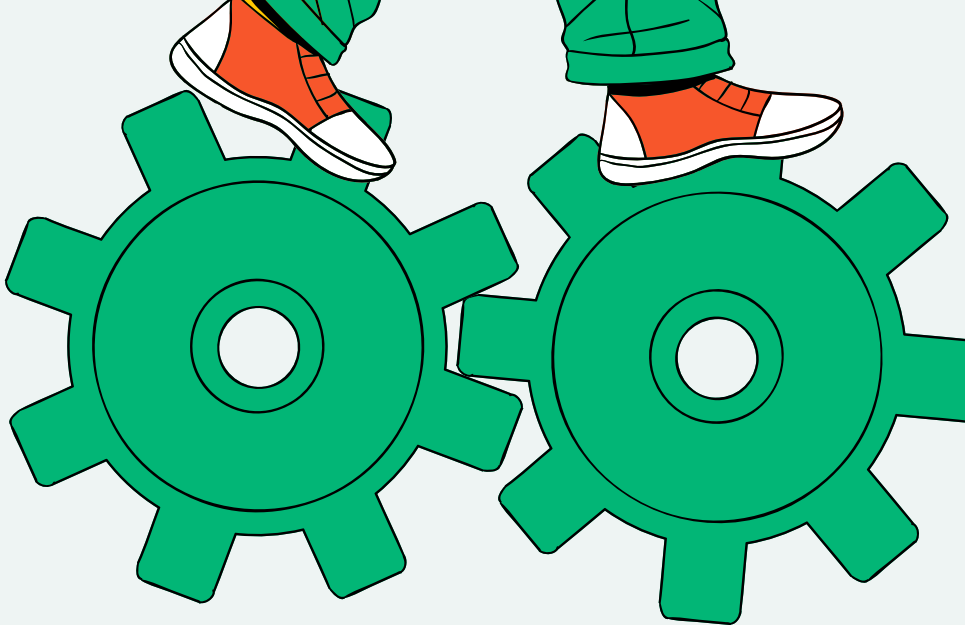


The table summarizes the performance of Apriori and FP-Growth algorithms under selected configurations. It shows runtime, number of rules, average confidence, average lift, and maximum lift for each setup. Apriori generates more rules and achieves higher lift but is slower, while FP-Growth is faster and scalable with fewer rules. This comparison highlights the trade-off between rule quality and computational efficiency, helping identify the best configuration for actionable insights.

Algorithm	Min Sup	Min Conf	Runtime (s)	# Rules	Avg Conf	Avg Lift	Max Lift
Apriori	0.001	0.1	15.63	1976	0.20	3.53	74.44
Apriori	0.002	0.3	4.35	54	0.35	11.49	74.44
Apriori	0.010	0.3	0.24	1	0.37	2.41	2.41
FP-Growth	0.005	0.1	8.17	112	0.20	2.14	5.18
FP-Growth	0.010	0.2	1.99	13	0.26	1.90	2.76



COMPARISON BETWEEN ALL THE MODELS WE BUILT



runs.summary["project_summary_numeric"]

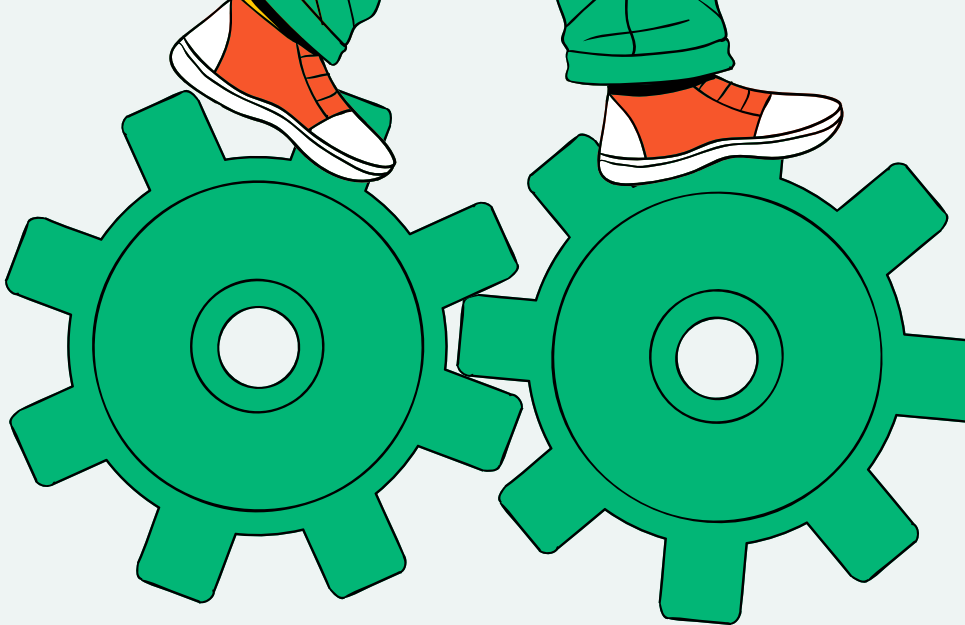
⚙

≡ Filter

	metric	value
1	total_experiments	16
2	fp_growth_count	4
3	apriori_count	12
	best_rules_count	54



COMPARAISON OF THE RULES GENERATED

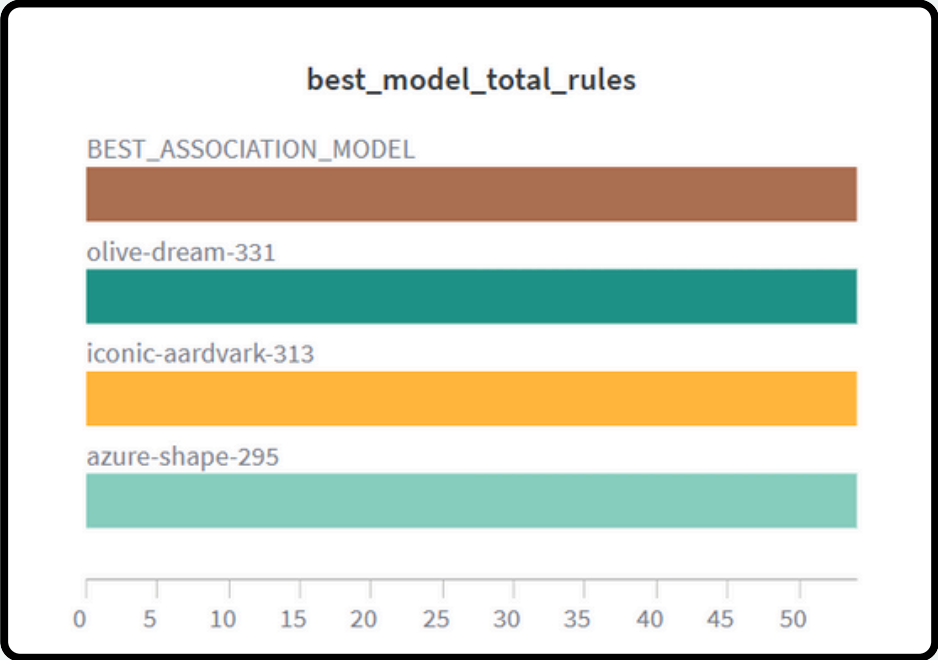
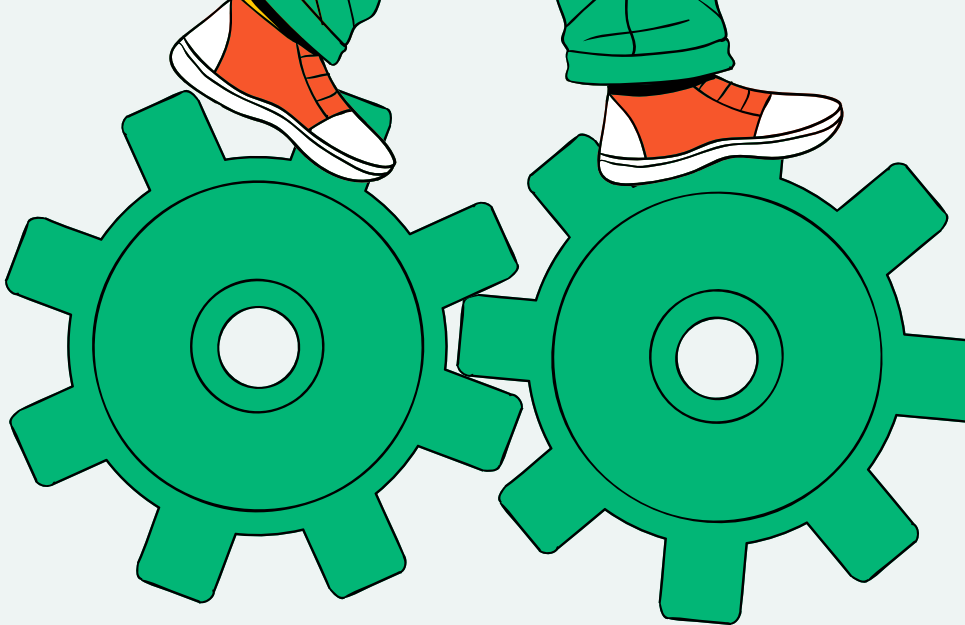


runs.summary["top_50_rules"]

	antecedents	consequents	antecedent su	consequent su	support	confidence	lift	representativit	leverage	conviction	zhangs_metric	jaccard	certainty
1	frozenset(Apple'})												
2	frozenset(Baby Spinach'})	frozenset(of Organic	0.07745	0.1234	0.0161	0.2079	1.685	1	0.006547	1.107	0.4408	0.08717	0.09642
3	frozenset(Hass Avocado'})	frozenset(of Organic	0.0706	0.1234	0.0195	0.2762	2.239	1	0.01079	1.211	0.5954	0.1118	0.1744
4	frozenset(Raspberrie	frozenset(of Organic	0.04585	0.1234	0.0136	0.2966	2.405	1	0.007944	1.246	0.6122	0.0874	0.1976
5	frozenset(Strawberri	frozenset(of Organic	0.08245	0.1234	0.0197	0.2389	1.937	1	0.00953	1.152	0.5272	0.1059	0.1318
6	frozenset(Lemon'})	frozenset(of Organic	0.05125	0.1544	0.0132	0.2576	1.668	1	0.005287	1.139	0.4222	0.06859	0.122
7	frozenset(of Organic	frozenset(of Organic	0.04635	0.1544	0.0109	0.2352	1.523	1	0.003744	1.106	0.3601	0.05741	0.09551



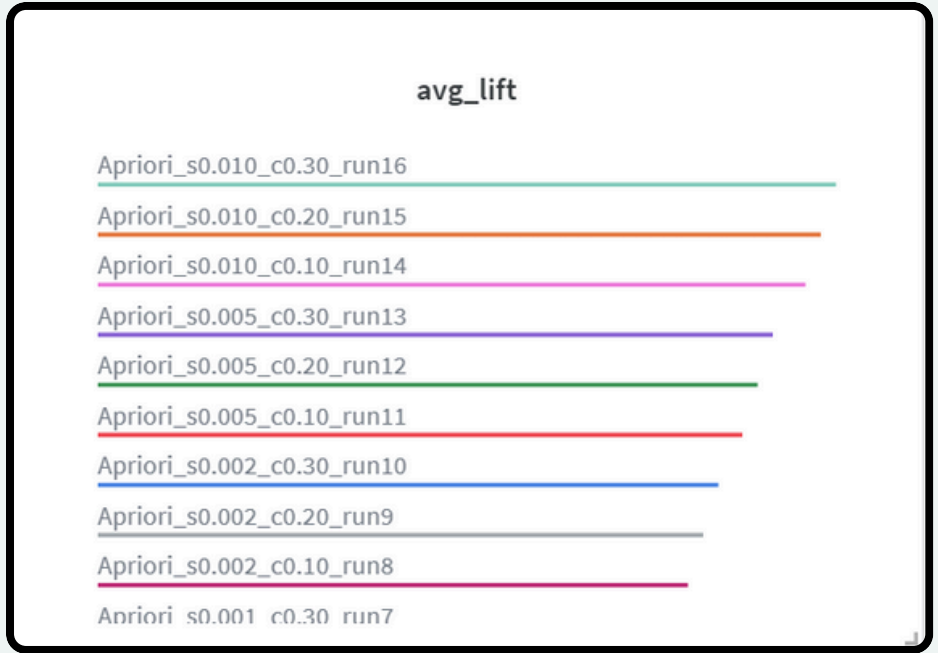
THE BEST MODEL COMPARAISON



The totale ruls of the best model



The efficiency score



The lift avrage

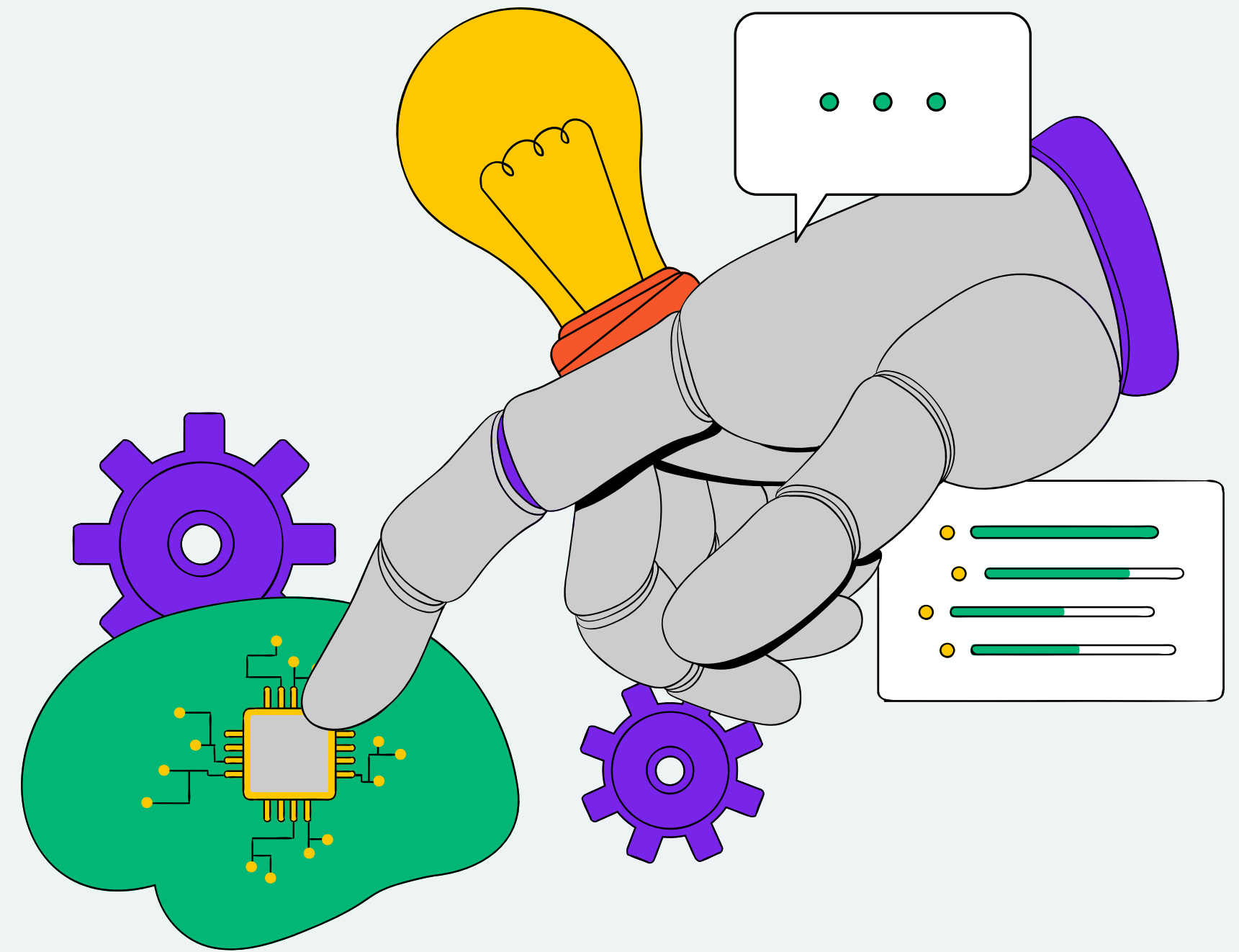


The confidence avrage



RESOURCES

The complete source code of this project is available on [GitHub](#), providing full access to the implementation and experiments. In addition, experiment tracking, model metrics, and training visualizations are documented using [Weights & Biases](#), ensuring transparency, reproducibility, and easy performance analysis throughout the project.



THANK YOU !

